

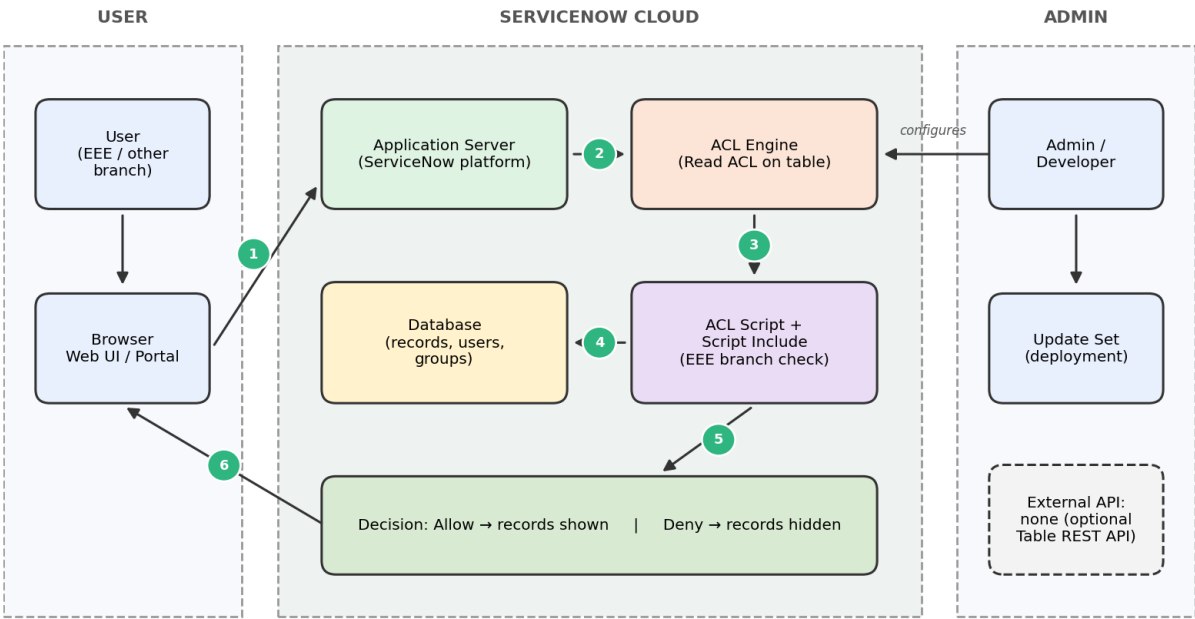
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	6ab221fbe880316a482c6241
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2

Example: Script-controlled ACL for restricting record access to the EEE branch



Guidelines:

- Include all the processes (as an application logic / technology block)
- Provide infrastructural demarcation (local / cloud)
- Indicate external interfaces (third party APIs etc.)
- Indicate data storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application, i.e. Web UI to open the records list and form	ServiceNow Next Experience UI / Service Portal, HTML, CSS, JavaScript (client scripts)
2.	Application Logic-1	Read ACL on the records table that decides whether a record can be viewed	ServiceNow Access Control (ACL), server-side JavaScript
3.	Application Logic-2	ACL script that checks whether the logged-in user belongs to the EEE branch (admin bypass allowed)	GlideSystem API: <code>gs.getUser().isMemberOf()</code> , <code>gs.hasRole()</code>
4.	Application Logic-3	Reusable branch-check logic that can be called from many ACLs	Script Include (JavaScript)

S.No	Component	Description	Technology
5.	Database	Stores records, users, groups and ACL configuration	ServiceNow relational database (custom records table, sys_user, sys_user_group)
6.	Cloud Database	Database service hosted on the cloud	ServiceNow cloud instance database
7.	File Storage	Attachments on records and Update Set export files	ServiceNow attachment storage, Update Set XML
8.	External API-1	Not required. Optional REST access to the table is protected by the same ACL	ServiceNow Table REST API (optional)
9.	External API-2	Not used in this project	N/A
10.	Machine Learning Model	Not used in this project	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on the cloud. Local Server Configuration: not required (browser only) Cloud Server Configuration: ServiceNow instance	Cloud – ServiceNow Personal Developer Instance (PDI)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No external open-source framework is used. The solution is built with platform-native scripting	ServiceNow platform JavaScript and Glide APIs
2.	Security Implementations	Access is controlled by ACLs with role and script conditions, group membership checks and admin bypass. Tested by impersonating EEE and non-EEE users	ServiceNow ACL, roles, groups, user authentication, HTTPS
3.	Scalable Architecture	Multi-tier cloud platform (client, application and database tiers). The same ACL pattern can be reused for other branches, tables and operations	ServiceNow multi-tier cloud architecture, Script Include
4.	Availability	The ServiceNow cloud manages availability, redundancy and backups, so no separate servers are needed. A Personal Developer Instance may hibernate when idle	ServiceNow cloud infrastructure
5.	Performance	The ACL script is kept short (a single group check) and avoids extra database queries, so record access stays fast	Lightweight ACL script, platform caching